

USART

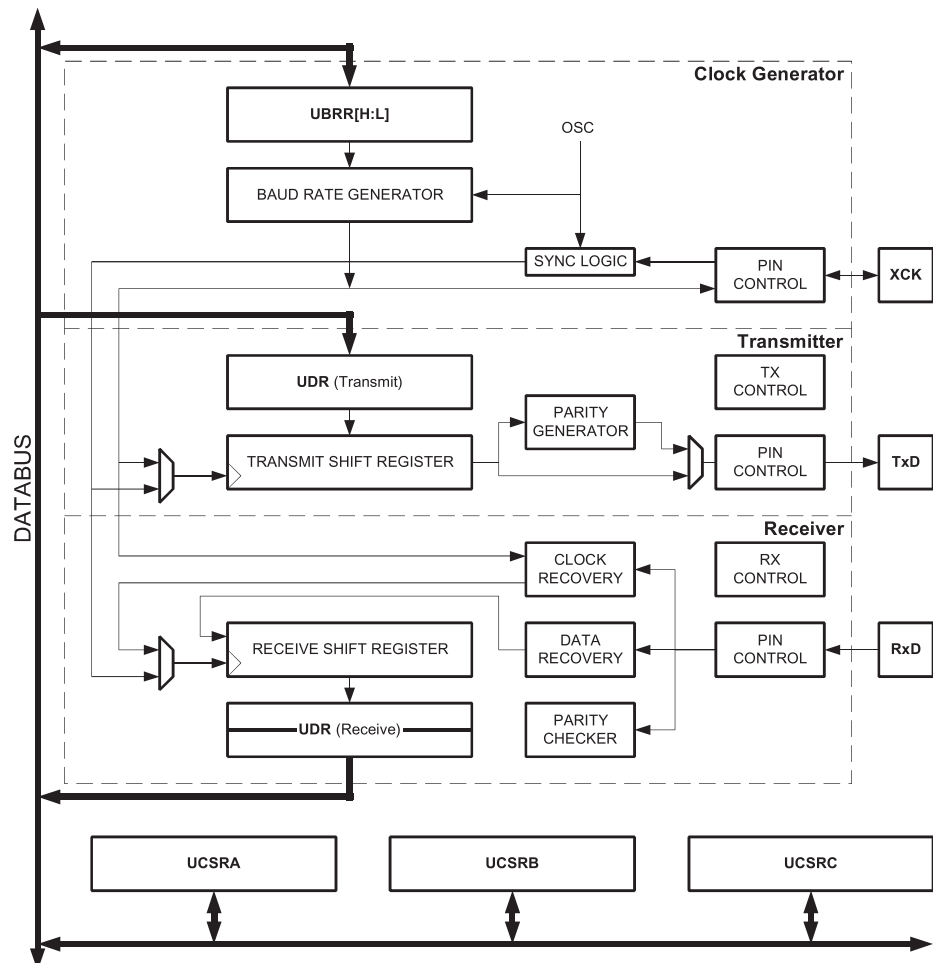
The Universal Synchronous and Asynchronous serial Receiver and Transmitter (USART) is a highly flexible serial communication device. The main features are:

- **Full Duplex Operation (Independent Serial Receive and Transmit Registers)**
- **Asynchronous or Synchronous Operation**
- **Master or Slave Clocked Synchronous Operation**
- **High Resolution Baud Rate Generator**
- **Supports Serial Frames with 5, 6, 7, 8, or 9 Data Bits and 1 or 2 Stop Bits**
- **Odd or Even Parity Generation and Parity Check Supported by Hardware**
- **Data OverRun Detection**
- **Framing Error Detection**
- **Noise Filtering Includes False Start Bit Detection and Digital Low Pass Filter**
- **Three Separate Interrupts on TX Complete, TX Data Register Empty, and RX Complete**
- **Multi-processor Communication Mode**
- **Double Speed Asynchronous Communication Mode**

Overview

A simplified block diagram of the USART transmitter is shown in [Figure 69](#). CPU accessible I/O Registers and I/O pins are shown in bold.

Figure 69. USART Block Diagram⁽¹⁾



Note: 1. Refer to [Figure 1](#) on page 2, [Table 33](#) on page 65, and [Table 27](#) on page 60 for USART pin placement.

The dashed boxes in the block diagram separate the three main parts of the USART (listed from the top): Clock Generator, Transmitter and Receiver. Control Registers are shared by all units. The clock generation logic consists of synchronization logic for external clock input used by synchronous Slave operation, and the baud rate generator. The XCK (Transfer Clock) pin is only used by Synchronous Transfer mode. The Transmitter consists of a single write buffer, a serial Shift Register, parity generator and control logic for handling different serial frame formats. The write buffer allows a continuous transfer of data without any delay between frames. The Receiver is the most complex part of the USART module due to its clock and data recovery units. The recovery units are used for asynchronous data reception. In addition to the recovery units, the receiver includes a parity checker, control logic, a Shift Register and a two level receive buffer (UDR). The receiver supports the same frame formats as the transmitter, and can detect frame error, data overrun and parity errors.

AVR USART vs. AVR UART – Compatibility

The USART is fully compatible with the AVR UART regarding:

- Bit locations inside all USART Registers
- Baud Rate Generation
- Transmitter Operation
- Transmit Buffer Functionality
- Receiver Operation

However, the receive buffering has two improvements that will affect the compatibility in some special cases:

- A second Buffer Register has been added. The two Buffer Registers operate as a circular FIFO buffer. Therefore the UDR must only be read once for each incoming data! More important is the fact that the Error Flags (FE and DOR) and the 9th data bit (RXB8) are buffered with the data in the receive buffer. Therefore the status bits must always be read before the UDR Register is read. Otherwise the error status will be lost since the buffer state is lost.
- The receiver Shift Register can now act as a third buffer level. This is done by allowing the received data to remain in the serial Shift Register (see [Figure 69](#)) if the Buffer Registers are full, until a new start bit is detected. The USART is therefore more resistant to Data OverRun (DOR) error conditions.

The following control bits have changed name, but have same functionality and register location:

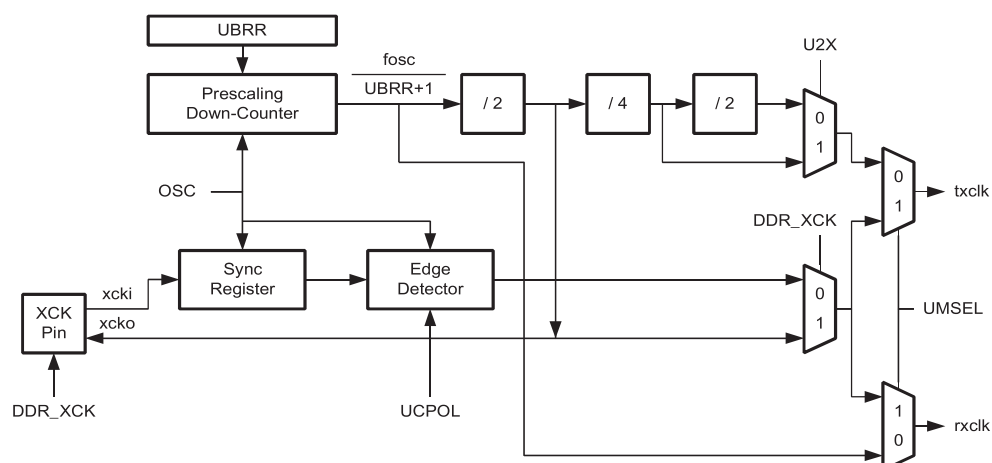
- CHR9 is changed to UCSZ2
- OR is changed to DOR

Clock Generation

The clock generation logic generates the base clock for the Transmitter and Receiver. The USART supports four modes of clock operation: Normal Asynchronous, Double Speed Asynchronous, Master Synchronous and Slave Synchronous mode. The UMSEL bit in USART Control and Status Register C (UCSRC) selects between asynchronous and synchronous operation. Double Speed (Asynchronous mode only) is controlled by the U2X found in the UCSRA Register. When using Synchronous mode (UMSEL = 1), the Data Direction Register for the XCK pin (DDR_XCK) controls whether the clock source is internal (Master mode) or external (Slave mode). The XCK pin is only active when using Synchronous mode.

[Figure 70](#) shows a block diagram of the clock generation logic.

Figure 70. Clock Generation Logic, Block Diagram



Signal description:

txclk Transmitter clock (Internal Signal).

rxclk Receiver base clock (Internal Signal).

xcki Input from XCK pin (Internal Signal). Used for synchronous Slave operation.

xcko Clock output to XCK pin (Internal Signal). Used for synchronous Master operation.

fosc XTAL pin frequency (System Clock).

Internal Clock Generation – The Baud Rate Generator

Internal clock generation is used for the asynchronous and the synchronous Master modes of operation. The description in this section refers to [Figure 70](#).

The USART Baud Rate Register (UBRR) and the down-counter connected to it function as a programmable prescaler or baud rate generator. The down-counter, running at system clock (fosc), is loaded with the UBRR value each time the counter has counted down to zero or when the UBRR Register is written. A clock is generated each time the counter reaches zero. This clock is the baud rate generator clock output ($= fosc/(UBRR+1)$). The Transmitter divides the baud rate generator clock output by 2, 8 or 16 depending on mode. The baud rate generator output is used directly by the receiver's clock and data recovery units. However, the recovery units use a state machine that uses 2, 8 or 16 states depending on mode set by the state of the UMSEL, U2X and DDR_XCK bits.

Table 60 contains equations for calculating the baud rate (in bits per second) and for calculating the UBRR value for each mode of operation using an internally generated clock source.

Table 60. Equations for Calculating Baud Rate Register Setting

Operating Mode	Equation for Calculating Baud Rate ⁽¹⁾	Equation for Calculating UBRR Value
Asynchronous Normal Mode (U2X = 0)	$BAUD = \frac{f_{OSC}}{16(UBRR + 1)}$	$UBRR = \frac{f_{OSC}}{16BAUD} - 1$
Asynchronous Double Speed Mode (U2X = 1)	$BAUD = \frac{f_{OSC}}{8(UBRR + 1)}$	$UBRR = \frac{f_{OSC}}{8BAUD} - 1$
Synchronous Master Mode	$BAUD = \frac{f_{OSC}}{2(UBRR + 1)}$	$UBRR = \frac{f_{OSC}}{2BAUD} - 1$

Note: 1. The baud rate is defined to be the transfer rate in bit per second (bps).

BAUD Baud rate (in bits per second, bps)

f_{OSC} System Oscillator clock frequency

UBRR Contents of the UBRRH and UBRL Registers, (0 - 4095)

Some examples of UBRR values for some system clock frequencies are found in [Table 68](#) (see [page 168](#)).

Double Speed Operation (U2X)

The transfer rate can be doubled by setting the U2X bit in UCSRA. Setting this bit only has effect for the asynchronous operation. Set this bit to zero when using synchronous operation.

Setting this bit will reduce the divisor of the baud rate divider from 16 to 8, effectively doubling the transfer rate for asynchronous communication. Note however that the receiver will in this case only use half the number of samples (reduced from 16 to 8) for data sampling and clock recovery, and therefore a more accurate baud rate setting and system clock are required when this mode is used. For the Transmitter, there are no downsides.

External Clock

External clocking is used by the synchronous Slave modes of operation. The description in this section refers to [Figure 70](#) for details.

External clock input from the XCK pin is sampled by a synchronization register to minimize the chance of meta-stability. The output from the synchronization register must then pass through an edge detector before it can be used by the Transmitter and receiver. This process introduces a two CPU clock period delay and therefore the maximum external XCK clock frequency is limited by the following equation:

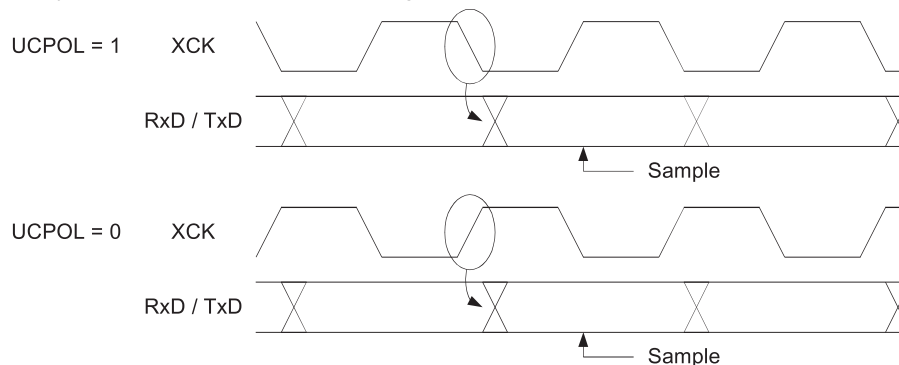
$$f_{XCK} < \frac{f_{OSC}}{4}$$

Note that f_{OSC} depends on the stability of the system clock source. It is therefore recommended to add some margin to avoid possible loss of data due to frequency variations.

Synchronous Clock Operation

When Synchronous mode is used (UMSEL = 1), the XCK pin will be used as either clock input (Slave) or clock output (Master). The dependency between the clock edges and data sampling or data change is the same. The basic principle is that data input (on RxD) is sampled at the opposite XCK clock edge of the edge the data output (TxD) is changed.

Figure 71. Synchronous Mode XCK Timing.



The UCPOL bit UCRSC selects which XCK clock edge is used for data sampling and which is used for data change. As [Figure 71](#) shows, when UCPOL is zero the data will be changed at rising XCK edge and sampled at falling XCK edge. If UCPOL is set, the data will be changed at falling XCK edge and sampled at rising XCK edge.

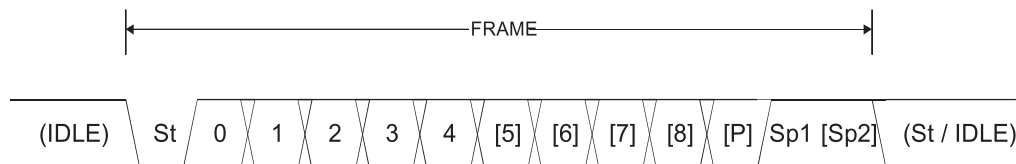
Frame Formats

A serial frame is defined to be one character of data bits with synchronization bits (start and stop bits), and optionally a parity bit for error checking. The USART accepts all 30 combinations of the following as valid frame formats:

- 1 start bit
- 5, 6, 7, 8, or 9 data bits
- no, even or odd parity bit
- 1 or 2 stop bits

A frame starts with the start bit followed by the least significant data bit. Then the next data bits, up to a total of nine, are succeeding, ending with the most significant bit. If enabled, the parity bit is inserted after the data bits, before the stop bits. When a complete frame is transmitted, it can be directly followed by a new frame, or the communication line can be set to an idle (high) state. [Figure 72](#) illustrates the possible combinations of the frame formats. Bits inside brackets are optional.

Figure 72. Frame Formats



- St** Start bit, always low.
- (n)** Data bits (0 to 8).
- P** Parity bit. Can be odd or even.
- Sp** Stop bit, always high.

IDLE No transfers on the communication line (RxD or TxD). An IDLE line must be high.

The frame format used by the USART is set by the UCSZ2:0, UPM1:0, and USBS bits in UCSRB and UCSRC. The Receiver and Transmitter use the same setting. Note that changing the setting of any of these bits will corrupt all ongoing communication for both the Receiver and Transmitter.

The USART Character SiZe (UCSZ2:0) bits select the number of data bits in the frame. The USART Parity mode (UPM1:0) bits enable and set the type of parity bit. The selection between one or two stop bits is done by the USART Stop Bit Select (USBS) bit. The receiver ignores the second stop bit. An FE (Frame Error) will therefore only be detected in the cases where the first stop bit is zero.

Parity Bit Calculation

The parity bit is calculated by doing an exclusive-or of all the data bits. If odd parity is used, the result of the exclusive or is inverted. The relation between the parity bit and data bits is as follows::

$$P_{even} = d_{n-1} \oplus \dots \oplus d_3 \oplus d_2 \oplus d_1 \oplus d_0 \oplus 0$$

$$P_{odd} = d_{n-1} \oplus \dots \oplus d_3 \oplus d_2 \oplus d_1 \oplus d_0 \oplus 1$$

P_{even} Parity bit using even parity

P_{odd} Parity bit using odd parity

d_n Data bit n of the character

If used, the parity bit is located between the last data bit and first stop bit of a serial frame.

USART Initialization

The USART has to be initialized before any communication can take place. The initialization process normally consists of setting the baud rate, setting frame format and enabling the Transmitter or the Receiver depending on the usage. For interrupt driven USART operation, the Global Interrupt Flag should be cleared (and interrupts globally disabled) when doing the initialization.

Before doing a re-initialization with changed baud rate or frame format, be sure that there are no ongoing transmissions during the period the registers are changed. The TXC Flag can be used to check that the Transmitter has completed all transfers, and the RXC Flag can be used to check that there are no unread data in the receive buffer. Note that the TXC Flag must be cleared before each transmission (before UDR is written) if it is used for this purpose.

The following simple USART initialization code examples show one assembly and one C function that are equal in functionality. The examples assume asynchronous operation using polling (no interrupts enabled) and a fixed frame format. The baud rate is given as a function parameter. For the assembly code, the baud rate parameter is assumed to be stored in the r17:r16 registers. When the function writes to the UCSRC Register, the URSEL bit (MSB) must be set due to the sharing of I/O location by UBRRH and UCSRC.

Assembly Code Example⁽¹⁾

```

USART_Init:
    ; Set baud rate
    out    UBRRH, r17
    out    UBRRL, r16
    ; Enable receiver and transmitter
    ldi    r16, (1<<RXEN) | (1<<TXEN)
    out    UCSRB, r16
    ; Set frame format: 8data, 2stop bit
    ldi    r16, (1<<URSEL) | (1<<USBS) | (3<<UCSZ0)
    out    UCSRC, r16
    ret

```

C Code Example⁽¹⁾

```

#define FOSC 1843200// Clock Speed
#define BAUD 9600
#define MYUBRR FOSC/16/BAUD-1
void main( void )
{
    ...
    USART_Init ( MYUBRR );
    ...
}

void USART_Init( unsigned int ubrr)
{
    /* Set baud rate */
    UBRRH = (unsigned char)(ubrr>>8);
    UBRRL = (unsigned char)ubrr;
    /* Enable receiver and transmitter */
    UCSRB = (1<<RXEN) | (1<<TXEN);
    /* Set frame format: 8data, 2stop bit */
    UCSRC = (1<<URSEL) | (1<<USBS) | (3<<UCSZ0);
}

```

Note: 1. See "About Code Examples" on page 7.

More advanced initialization routines can be made that include frame format as parameters, disable interrupts and so on. However, many applications use a fixed setting of the Baud and Control Registers, and for these types of applications the initialization code can be placed directly in the main routine, or be combined with initialization code for other I/O modules.

Data Transmission – The USART Transmitter

The USART Transmitter is enabled by setting the *Transmit Enable* (TXEN) bit in the UCSRB Register. When the Transmitter is enabled, the normal port operation of the TxD pin is overridden by the USART and given the function as the transmitter's serial output. The baud rate, mode of operation and frame format must be set up once before doing any transmissions. If synchronous operation is used, the clock on the XCK pin will be overridden and used as transmission clock.

Sending Frames with 5 to 8 Data Bit

A data transmission is initiated by loading the transmit buffer with the data to be transmitted. The CPU can load the transmit buffer by writing to the UDR I/O location. The buffered data in the transmit buffer will be moved to the Shift Register when the Shift Register is ready to send a new

frame. The Shift Register is loaded with new data if it is in idle state (no ongoing transmission) or immediately after the last stop bit of the previous frame is transmitted. When the Shift Register is loaded with new data, it will transfer one complete frame at the rate given by the Baud Register, U2X bit or by XCK depending on mode of operation.

The following code examples show a simple USART transmit function based on polling of the *Data Register Empty* (UDRE) Flag. When using frames with less than eight bits, the most significant bits written to the UDR are ignored. The USART has to be initialized before the function can be used. For the assembly code, the data to be sent is assumed to be stored in Register R16

Assembly Code Example⁽¹⁾

```
USART_Transmit:
    ; Wait for empty transmit buffer
    sbis UCSRA,UDRE
    rjmp USART_Transmit
    ; Put data (r16) into buffer, sends the data
    out  UDR,r16
    ret
```

C Code Example⁽¹⁾

```
void USART_Transmit( unsigned char data )
{
    /* Wait for empty transmit buffer */
    while ( !( UCSRA & (1<<UDRE)) )
        ;
    /* Put data into buffer, sends the data */
    UDR = data;
}
```

Note: 1. See "About Code Examples" on page 7.

The function simply waits for the transmit buffer to be empty by checking the UDRE Flag, before loading it with new data to be transmitted. If the Data Register Empty Interrupt is utilized, the interrupt routine writes the data into the buffer.

Sending Frames with 9 Data Bit

If 9-bit characters are used (UCSZ = 7), the ninth bit must be written to the TXB8 bit in UCSRB before the Low byte of the character is written to UDR. The following code examples show a transmit function that handles 9-bit characters. For the assembly code, the data to be sent is assumed to be stored in Registers R17:R16.

Assembly Code Example⁽¹⁾

```

USART_Transmit:
    ; Wait for empty transmit buffer
    sbis UCSRA,UDRE
    rjmp USART_Transmit
    ; Copy 9th bit from r17 to TXB8
    cbi UCSRB,TXB8
    sbrc r17,0
    sbi UCSRB,TXB8
    ; Put LSB data (r16) into buffer, sends the data
    out UDR,r16
    ret
    
```

C Code Example⁽¹⁾

```

void USART_Transmit( unsigned int data )
{
    /* Wait for empty transmit buffer */
    while ( !( UCSRA & (1<<UDRE)) )
        ;

    /* Copy 9th bit to TXB8 */
    UCSRB &= ~(1<<TXB8);
    if ( data & 0x0100 )
        UCSRB |= (1<<TXB8);
    /* Put data into buffer, sends the data */
    UDR = data;
}
    
```

Note: 1. These transmit functions are written to be general functions. They can be optimized if the contents of the UCSRB is static. (that is, only the TXB8 bit of the UCSRB Register is used after initialization).

The ninth bit can be used for indicating an address frame when using multi processor communication mode or for other protocol handling as for example synchronization.

Transmitter Flags and Interrupts

The USART transmitter has two flags that indicate its state: USART Data Register Empty (UDRE) and Transmit Complete (TXC). Both flags can be used for generating interrupts.

The Data Register Empty (UDRE) Flag indicates whether the transmit buffer is ready to receive new data. This bit is set when the transmit buffer is empty, and cleared when the transmit buffer contains data to be transmitted that has not yet been moved into the Shift Register. For compatibility with future devices, always write this bit to zero when writing the UCSRA Register.

When the Data Register empty Interrupt Enable (UDRIE) bit in UCSRB is written to one, the USART Data Register Empty Interrupt will be executed as long as UDRE is set (provided that global interrupts are enabled). UDRE is cleared by writing UDR. When interrupt-driven data transmission is used, the Data Register Empty Interrupt routine must either write new data to UDR in order to clear UDRE or disable the Data Register empty Interrupt, otherwise a new interrupt will occur once the interrupt routine terminates.

The Transmit Complete (TXC) Flag bit is set one when the entire frame in the transmit Shift Register has been shifted out and there are no new data currently present in the transmit buffer. The TXC Flag bit is automatically cleared when a transmit complete interrupt is executed, or it can be cleared by writing a one to its bit location. The TXC Flag is useful in half-duplex communication interfaces (like the RS485 standard), where a transmitting application must enter receive mode and free the communication bus immediately after completing the transmission.

When the Transmit Complete Interrupt Enable (TXCIE) bit in UCSRB is set, the USART Transmit Complete Interrupt will be executed when the TXC Flag becomes set (provided that global interrupts are enabled). When the transmit complete interrupt is used, the interrupt handling routine does not have to clear the TXC Flag, this is done automatically when the interrupt is executed.

Parity Generator

The parity generator calculates the parity bit for the serial frame data. When parity bit is enabled (UPM1 = 1), the transmitter control logic inserts the parity bit between the last data bit and the first stop bit of the frame that is sent.

Disabling the Transmitter

The disabling of the transmitter (setting the TXEN to zero) will not become effective until ongoing and pending transmissions are completed, that is, when the transmit Shift Register and transmit Buffer Register do not contain data to be transmitted. When disabled, the transmitter will no longer override the TxD pin.

Data Reception – The USART Receiver

The USART Receiver is enabled by writing the Receive Enable (RXEN) bit in the UCSRB Register to one. When the receiver is enabled, the normal pin operation of the RxD pin is overridden by the USART and given the function as the receiver's serial input. The baud rate, mode of operation and frame format must be set up once before any serial reception can be done. If synchronous operation is used, the clock on the XCK pin will be used as transfer clock.

Receiving Frames with 5 to 8 Data Bits

The receiver starts data reception when it detects a valid start bit. Each bit that follows the start bit will be sampled at the baud rate or XCK clock, and shifted into the receive Shift Register until the first stop bit of a frame is received. A second stop bit will be ignored by the receiver. When the first stop bit is received, that is, a complete serial frame is present in the receive Shift Register, the contents of the Shift Register will be moved into the receive buffer. The receive buffer can then be read by reading the UDR I/O location.

The following code example shows a simple USART receive function based on polling of the Receive Complete (RXC) Flag. When using frames with less than eight bits the most significant bits of the data read from the UDR will be masked to zero. The USART has to be initialized before the function can be used.

Assembly Code Example⁽¹⁾

```
USART_Receive:
    ; Wait for data to be received
    sbis UCSRA, RXC
    rjmp USART_Receive
    ; Get and return received data from buffer
    in    r16, UDR
    ret
```

C Code Example⁽¹⁾

```
unsigned char USART_Receive( void )
{
    /* Wait for data to be received */
    while ( !(UCSRA & (1<<RXC)) )
        ;
    /* Get and return received data from buffer */
    return UDR;
}
```

Note: 1. See "About Code Examples" on page 7.

The function simply waits for data to be present in the receive buffer by checking the RXC Flag, before reading the buffer and returning the value.

Receiving Frames with 9 Databits

If 9 bit characters are used (UCSZ=7) the ninth bit must be read from the RXB8 bit in UCSRB **before** reading the low bits from the UDR. This rule applies to the FE, DOR and PE status Flags as well. Read status from UCSRA, then data from UDR. Reading the UDR I/O location will change the state of the receive buffer FIFO and consequently the TXB8, FE, DOR and PE bits, which all are stored in the FIFO, will change.

The following code example shows a simple USART receive function that handles both 9-bit characters and the status bits.

Assembly Code Example⁽¹⁾

```

USART_Receive:
    ; Wait for data to be received
    sbis UCSRA, RXC
    rjmp USART_Receive
    ; Get status and 9th bit, then data from buffer
    in    r18, UCSRA
    in    r17, UCSRB
    in    r16, UDR
    ; If error, return -1
    andi  r18, (1<<FE) | (1<<DOR) | (1<<PE)
    breq  USART_ReceiveNoError
    ldi   r17, HIGH(-1)
    ldi   r16, LOW(-1)
USART_ReceiveNoError:
    ; Filter the 9th bit, then return
    lsr   r17
    andi  r17, 0x01
    ret
    
```

C Code Example⁽¹⁾

```

unsigned int USART_Receive( void )
{
    unsigned char status, resh, resl;
    /* Wait for data to be received */
    while ( !(UCSRA & (1<<RXC)) )
        ;
    /* Get status and 9th bit, then data */
    /* from buffer */
    status = UCSRA;
    resh = UCSRB;
    resl = UDR;
    /* If error, return -1 */
    if ( status & (1<<FE) | (1<<DOR) | (1<<PE) )
        return -1;
    /* Filter the 9th bit, then return */
    resh = (resh >> 1) & 0x01;
    return ((resh << 8) | resl);
}
    
```

Note: 1. See "About Code Examples" on page 7.

The receive function example reads all the I/O Registers into the Register File before any computation is done. This gives an optimal receive buffer utilization since the buffer location read will be free to accept new data as early as possible.

Receive Complete Flag and Interrupt

The USART Receiver has one flag that indicates the receiver state.

The Receive Complete (RXC) Flag indicates if there are unread data present in the receive buffer. This flag is one when unread data exist in the receive buffer, and zero when the receive buffer is empty (that is, does not contain any unread data). If the receiver is disabled (RXEN = 0), the receive buffer will be flushed and consequently the RXC bit will become zero.

When the Receive Complete Interrupt Enable (RXCIE) in UCSRB is set, the USART Receive Complete Interrupt will be executed as long as the RXC Flag is set (provided that global interrupts are enabled). When interrupt-driven data reception is used, the receive complete routine must read the received data from UDR in order to clear the RXC Flag, otherwise a new interrupt will occur once the interrupt routine terminates.

Receiver Error Flags

The USART Receiver has three Error Flags: Frame Error (FE), Data OverRun (DOR) and Parity Error (PE). All can be accessed by reading UCSRA. Common for the Error Flags is that they are located in the receive buffer together with the frame for which they indicate the error status. Due to the buffering of the Error Flags, the UCSRA must be read before the receive buffer (UDR), since reading the UDR I/O location changes the buffer read location. Another equality for the Error Flags is that they can not be altered by software doing a write to the flag location. However, all flags must be set to zero when the UCSRA is written for upward compatibility of future USART implementations. None of the Error Flags can generate interrupts.

The Frame Error (FE) Flag indicates the state of the first stop bit of the next readable frame stored in the receive buffer. The FE Flag is zero when the stop bit was correctly read (as one), and the FE Flag will be one when the stop bit was incorrect (zero). This flag can be used for detecting out-of-sync conditions, detecting break conditions and protocol handling. The FE Flag is not affected by the setting of the USBF bit in UCSRC since the receiver ignores all, except for the first, stop bits. For compatibility with future devices, always set this bit to zero when writing to UCSRA.

The Data OverRun (DOR) Flag indicates data loss due to a receiver buffer full condition. A Data OverRun occurs when the receive buffer is full (two characters), it is a new character waiting in the receive Shift Register, and a new start bit is detected. If the DOR Flag is set there was one or more serial frame lost between the frame last read from UDR, and the next frame read from UDR. For compatibility with future devices, always write this bit to zero when writing to UCSRA. The DOR Flag is cleared when the frame received was successfully moved from the Shift Register to the receive buffer.

The Parity Error (PE) Flag indicates that the next frame in the receive buffer had a parity error when received. If parity check is not enabled the PE bit will always be read zero. For compatibility with future devices, always set this bit to zero when writing to UCSRA. For more details see [“Parity Bit Calculation” on page 149](#) and [“Parity Checker” on page 156](#).

Parity Checker

The Parity Checker is active when the high USART Parity mode (UPM1) bit is set. Type of parity check to be performed (odd or even) is selected by the UPM0 bit. When enabled, the parity checker calculates the parity of the data bits in incoming frames and compares the result with the parity bit from the serial frame. The result of the check is stored in the receive buffer together with the received data and stop bits. The Parity Error (PE) Flag can then be read by software to check if the frame had a parity error.

The PE bit is set if the next character that can be read from the receive buffer had a parity error when received and the parity checking was enabled at that point (UPM1 = 1). This bit is valid until the receive buffer (UDR) is read.

Disabling the Receiver In contrast to the Transmitter, disabling of the Receiver will be immediate. Data from ongoing receptions will therefore be lost. When disabled (that is, the RXEN is set to zero) the Receiver will no longer override the normal function of the RxD port pin. The receiver buffer FIFO will be flushed when the receiver is disabled. Remaining data in the buffer will be lost

Flushing the Receive Buffer The receiver buffer FIFO will be flushed when the Receiver is disabled, that is, the buffer will be emptied of its contents. Unread data will be lost. If the buffer has to be flushed during normal operation, due to for instance an error condition, read the UDR I/O location until the RXC Flag is cleared. The following code example shows how to flush the receive buffer.

Assembly Code Example⁽¹⁾

```
USART_Flush:
    sbis UCSRA, RXC
    ret
    in    r16, UDR
    rjmp  USART_Flush
```

C Code Example⁽¹⁾

```
void USART_Flush( void )
{
    unsigned char dummy;
    while ( UCSRA & (1<<RXC) ) dummy = UDR;
}
```

Note: 1. See "About Code Examples" on page 7.

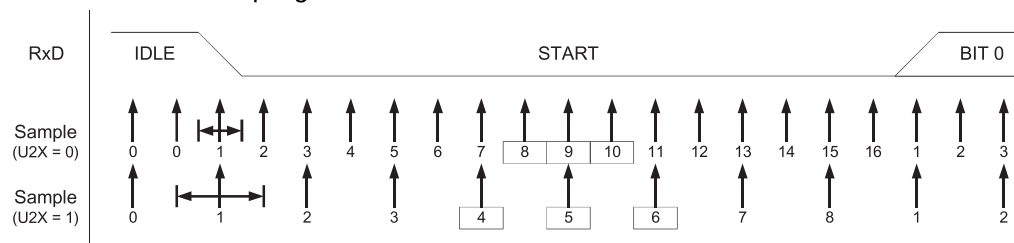
Asynchronous Data Reception

The USART includes a clock recovery and a data recovery unit for handling asynchronous data reception. The clock recovery logic is used for synchronizing the internally generated baud rate clock to the incoming asynchronous serial frames at the RxD pin. The data recovery logic samples and low pass filters each incoming bit, thereby improving the noise immunity of the receiver. The asynchronous reception operational range depends on the accuracy of the internal baud rate clock, the rate of the incoming frames, and the frame size in number of bits.

Asynchronous Clock Recovery

The clock recovery logic synchronizes internal clock to the incoming serial frames. [Figure 73](#) illustrates the sampling process of the start bit of an incoming frame. The sample rate is 16 times the baud rate for Normal mode, and 8 times the baud rate for Double Speed mode. The horizontal arrows illustrate the synchronization variation due to the sampling process. Note the larger time variation when using the double speed mode (U2X = 1) of operation. Samples denoted zero are samples done when the RxD line is idle (that is, no communication activity).

Figure 73. Start Bit Sampling



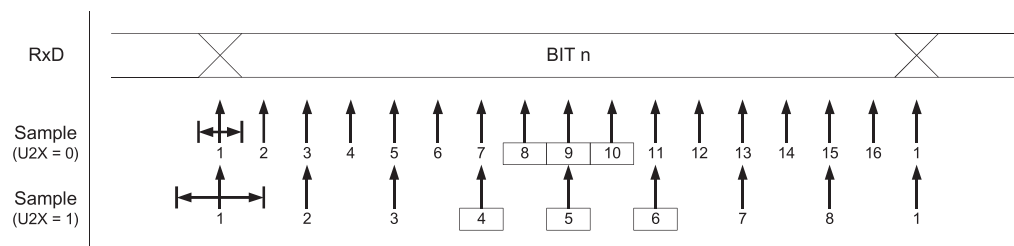
When the clock recovery logic detects a high (idle) to low (start) transition on the RxD line, the start bit detection sequence is initiated. Let sample 1 denote the first zero-sample as shown in the figure. The clock recovery logic then uses samples 8, 9, and 10 for Normal mode, and samples 4, 5, and 6 for Double Speed mode (indicated with sample numbers inside boxes on the

Asynchronous Data Recovery

figure), to decide if a valid start bit is received. If two or more of these three samples have logical high levels (the majority wins), the start bit is rejected as a noise spike and the receiver starts looking for the next high to low-transition. If however, a valid start bit is detected, the clock recovery logic is synchronized and the data recovery can begin. The synchronization process is repeated for each start bit.

When the receiver clock is synchronized to the start bit, the data recovery can begin. The data recovery unit uses a state machine that has 16 states for each bit in normal mode and 8 states for each bit in Double Speed mode. Figure 74 shows the sampling of the data bits and the parity bit. Each of the samples is given a number that is equal to the state of the recovery unit.

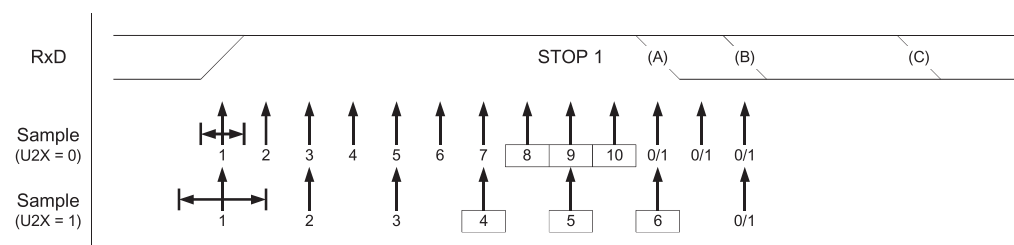
Figure 74. Sampling of Data and Parity Bit



The decision of the logic level of the received bit is taken by doing a majority voting of the logic value to the three samples in the center of the received bit. The center samples are emphasized on the figure by having the sample number inside boxes. The majority voting process is done as follows: If two or all three samples have high levels, the received bit is registered to be a logic 1. If two or all three samples have low levels, the received bit is registered to be a logic 0. This majority voting process acts as a low pass filter for the incoming signal on the RxD pin. The recovery process is then repeated until a complete frame is received. Including the first stop bit. Note that the receiver only uses the first stop bit of a frame.

Figure 75 shows the sampling of the stop bit and the earliest possible beginning of the start bit of the next frame.

Figure 75. Stop Bit Sampling and Next Start Bit Sampling



The same majority voting is done to the stop bit as done for the other bits in the frame. If the stop bit is registered to have a logic 0 value, the Frame Error (FE) Flag will be set.

A new high to low transition indicating the start bit of a new frame can come right after the last of the bits used for majority voting. For Normal Speed mode, the first low level sample can be at point marked (A) in Figure 75. For Double Speed mode the first low level must be delayed to (B). (C) marks a stop bit of full length. The early start bit detection influences the operational range of the receiver.

Asynchronous Operational Range

The operational range of the receiver is dependent on the mismatch between the received bit rate and the internally generated baud rate. If the Transmitter is sending frames at too fast or too slow bit rates, or the internally generated baud rate of the receiver does not have a similar (see [Table 61](#)) base frequency, the receiver will not be able to synchronize the frames to the start bit.

The following equations can be used to calculate the ratio of the incoming data rate and internal receiver baud rate.

$$R_{slow} = \frac{(D+1)S}{S-1+D \cdot S+S_F}$$

$$R_{fast} = \frac{(D+2)S}{(D+1)S+S_M}$$

- D Sum of character size and parity size (D = 5 to 10 bit)
- S Samples per bit. S = 16 for Normal Speed mode and S = 8 for Double Speed mode.
- S_F First sample number used for majority voting. S_F = 8 for Normal Speed and S_F = 4 for Double Speed mode.
- S_M Middle sample number used for majority voting. S_M = 9 for Normal Speed and S_M = 5 for Double Speed mode.
- R_{slow} is the ratio of the slowest incoming data rate that can be accepted in relation to the receiver baud rate. R_{fast} is the ratio of the fastest incoming data rate that can be accepted in relation to the receiver baud rate.

[Table 61](#) and [Table 62](#) list the maximum receiver baud rate error that can be tolerated. Note that Normal Speed mode has higher toleration of baud rate variations.

Table 61. Recommended Maximum Receiver Baud Rate Error for Normal Speed Mode (U2X = 0)

D # (Data+Parity Bit)	R _{slow} (%)	R _{fast} (%)	Max Total Error (%)	Recommended Max Receiver Error (%)
5	93.20	106.67	+6.67/-6.8	±3.0
6	94.12	105.79	+5.79/-5.88	±2.5
7	94.81	105.11	+5.11/-5.19	±2.0
8	95.36	104.58	+4.58/-4.54	±2.0
9	95.81	104.14	+4.14/-4.19	±1.5
10	96.17	103.78	+3.78/-3.83	±1.5

Table 62. Recommended Maximum Receiver Baud Rate Error for Double Speed Mode (U2X = 1)

D # (Data+Parity Bit)	R _{slow} (%)	R _{fast} (%)	Max Total Error (%)	Recommended Max Receiver Error (%)
5	94.12	105.66	+5.66/-5.88	±2.5
6	94.92	104.92	+4.92/-5.08	±2.0
7	95.52	104.35	+4.35/-4.48	±1.5

Table 62. Recommended Maximum Receiver Baud Rate Error for Double Speed Mode (U2X = 1)

D # (Data+Parity Bit)	R _{slow} (%)	R _{fast} (%)	Max Total Error (%)	Recommended Max Receiver Error (%)
8	96.00	103.90	+3.90/-4.00	±1.5
9	96.39	103.53	+3.53/-3.61	±1.5
10	96.70	103.23	+3.23/-3.30	±1.0

The recommendations of the maximum receiver baud rate error was made under the assumption that the receiver and transmitter equally divides the maximum total error.

There are two possible sources for the receivers baud rate error. The receiver's system clock (XTAL) will always have some minor instability over the supply voltage range and the temperature range. When using a crystal to generate the system clock, this is rarely a problem, but for a resonator the system clock may differ more than 2% depending of the resonators tolerance. The second source for the error is more controllable. The baud rate generator can not always do an exact division of the system frequency to get the baud rate wanted. In this case an UBRR value that gives an acceptable low error can be used if possible.

Multi-processor Communication Mode

Setting the Multi-processor Communication mode (MPCM) bit in UCSRA enables a filtering function of incoming frames received by the USART Receiver. Frames that do not contain address information will be ignored and not put into the receive buffer. This effectively reduces the number of incoming frames that has to be handled by the CPU, in a system with multiple MCUs that communicate via the same serial bus. The Transmitter is unaffected by the MPCM setting, but has to be used differently when it is a part of a system utilizing the Multi-processor Communication mode.

If the receiver is set up to receive frames that contain 5 to 8 data bits, then the first stop bit indicates if the frame contains data or address information. If the receiver is set up for frames with nine data bits, then the ninth bit (RXB8) is used for identifying address and data frames. When the frame type bit (the first stop or the ninth bit) is one, the frame contains an address. When the frame type bit is zero the frame is a data frame.

The Multi-processor Communication mode enables several Slave MCUs to receive data from a Master MCU. This is done by first decoding an address frame to find out which MCU has been addressed. If a particular Slave MCU has been addressed, it will receive the following data frames as normal, while the other Slave MCUs will ignore the received frames until another address frame is received.

Using MPCM

For an MCU to act as a Master MCU, it can use a 9-bit character frame format (UCSZ = 7). The ninth bit (TXB8) must be set when an address frame (TXB8 = 1) or cleared when a data frame (TXB8 = 0) is being transmitted. The Slave MCUs must in this case be set to use a 9-bit character frame format.

The following procedure should be used to exchange data in Multi-processor Communication mode:

1. All Slave MCUs are in Multi-processor Communication mode (MPCM in UCSRA is set).
2. The Master MCU sends an address frame, and all Slaves receive and read this frame. In the Slave MCUs, the RXC Flag in UCSRA will be set as normal.
3. Each Slave MCU reads the UDR Register and determines if it has been selected. If so, it clears the MPCM bit in UCSRA, otherwise it waits for the next address byte and keeps the MPCM setting.
4. The addressed MCU will receive all data frames until a new address frame is received. The other Slave MCUs, which still have the MPCM bit set, will ignore the data frames.
5. When the last data frame is received by the addressed MCU, the addressed MCU sets the MPCM bit and waits for a new address frame from Master. The process then repeats from 2.

Using any of the 5-bit to 8-bit character frame formats is possible, but impractical since the receiver must change between using n and $n+1$ character frame formats. This makes full-duplex operation difficult since the transmitter and receiver uses the same character size setting. If 5-bit to 8-bit character frames are used, the transmitter must be set to use two stop bit (USBS = 1) since the first stop bit is used for indicating the frame type.

Do not use Read-Modify-Write instructions (SBI and CBI) to set or clear the MPCM bit. The MPCM bit shares the same I/O location as the TXC Flag and this might accidentally be cleared when using SBI or CBI instructions.

Accessing UBRRH/ UCSRC Registers

The UBRRH Register shares the same I/O location as the UCSRC Register. Therefore some special consideration must be taken when accessing this I/O location.

Write Access

When doing a write access of this I/O location, the high bit of the value written, the USART Register Select (URSEL) bit, controls which one of the two registers that will be written. If URSEL is zero during a write operation, the UBRRH value will be updated. If URSEL is one, the UCSRC setting will be updated.

The following code examples show how to access the two registers.

Assembly Code Example⁽¹⁾

```
...
; Set UBRRH to 2
ldi r16,0x02
out UBRRH,r16
...
; Set the USBS and the UCSZ1 bit to one, and
; the remaining bits to zero.
ldi r16, (1<<URSEL) | (1<<USBS) | (1<<UCSZ1)
out UCSRC,r16
...
```

C Code Example⁽¹⁾

```
...
/* Set UBRRH to 2 */
UBRRH = 0x02;
...
/* Set the USBS and the UCSZ1 bit to one, and */
/* the remaining bits to zero. */
UCSRC = (1<<URSEL) | (1<<USBS) | (1<<UCSZ1);
...
```

Note: 1. See "About Code Examples" on page 7.

As the code examples illustrate, write accesses of the two registers are relatively unaffected of the sharing of I/O location.

Read Access

Doing a read access to the UBRRH or the UCSRC Register is a more complex operation. However, in most applications, it is rarely necessary to read any of these registers.

The read access is controlled by a timed sequence. Reading the I/O location once returns the UBRRH Register contents. If the register location was read in previous system clock cycle, reading the register in the current clock cycle will return the UCSRC contents. Note that the timed sequence for reading the UCSRC is an atomic operation. Interrupts must therefore be controlled (for example by disabling interrupts globally) during the read operation.

The following code example shows how to read the UCSRC Register contents.

Assembly Code Example⁽¹⁾

```
USART_ReadUCSRC:
    ; Read UCSRC
    in r16,UBRRH
    in r16,UCSRC
    ret
```

C Code Example⁽¹⁾

```
unsigned char USART_ReadUCSRC( void )
{
    unsigned char ucsrc;
    /* Read UCSRC */
    ucsrc = UBRRH;
    ucsrc = UCSRC;
    return ucsrc;
}
```

Note: 1. See “About Code Examples” on page 7.

The assembly code example returns the UCSRC value in r16.

Reading the UBRRH contents is not an atomic operation and therefore it can be read as an ordinary register, as long as the previous instruction did not access the register location.

USART Register Description

USART I/O Data Register – UDR

Bit	7	6	5	4	3	2	1	0	
	RXB[7:0]								UDR (Read)
	TXB[7:0]								UDR (Write)
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

The USART Transmit Data Buffer Register and USART Receive Data Buffer Registers share the same I/O address referred to as USART Data Register or UDR. The Transmit Data Buffer Register (TXB) will be the destination for data written to the UDR Register location. Reading the UDR Register location will return the contents of the Receive Data Buffer Register (RXB).

For 5-bit, 6-bit, or 7-bit characters the upper unused bits will be ignored by the Transmitter and set to zero by the Receiver.

The transmit buffer can only be written when the UDRE Flag in the UCSRA Register is set. Data written to UDR when the UDRE Flag is not set, will be ignored by the USART Transmitter. When data is written to the transmit buffer, and the Transmitter is enabled, the Transmitter will load the data into the transmit Shift Register when the Shift Register is empty. Then the data will be serially transmitted on the TxD pin.

The receive buffer consists of a two level FIFO. The FIFO will change its state whenever the receive buffer is accessed. Due to this behavior of the receive buffer, do not use read modify write instructions (SBI and CBI) on this location. Be careful when using bit test instructions (SBIC and SBIS), since these also will change the state of the FIFO.

USART Control and Status Register A – UCSRA

Bit	7	6	5	4	3	2	1	0	
	RXC	TXC	UDRE	FE	DOR	PE	U2X	MPCM	UCSRA
Read/Write	R	R/W	R	R	R	R	R/W	R/W	
Initial Value	0	0	1	0	0	0	0	0	

• Bit 7 – RXC: USART Receive Complete

This flag bit is set when there are unread data in the receive buffer and cleared when the receive buffer is empty (that is, does not contain any unread data). If the receiver is disabled, the receive buffer will be flushed and consequently the RXC bit will become zero. The RXC Flag can be used to generate a Receive Complete interrupt (see description of the RXCIE bit).

• Bit 6 – TXC: USART Transmit Complete

This flag bit is set when the entire frame in the transmit Shift Register has been shifted out and there are no new data currently present in the transmit buffer (UDR). The TXC Flag bit is automatically cleared when a transmit complete interrupt is executed, or it can be cleared by writing a one to its bit location. The TXC Flag can generate a Transmit Complete interrupt (see description of the TXCIE bit).

• Bit 5 – UDRE: USART Data Register Empty

The UDRE Flag indicates if the transmit buffer (UDR) is ready to receive new data. If UDRE is one, the buffer is empty, and therefore ready to be written. The UDRE Flag can generate a Data Register empty Interrupt (see description of the UDRIE bit).

UDRE is set after a reset to indicate that the transmitter is ready.

• Bit 4 – FE: Frame Error

This bit is set if the next character in the receive buffer had a Frame Error when received. that is, when the first stop bit of the next character in the receive buffer is zero. This bit is valid until the receive buffer (UDR) is read. The FE bit is zero when the stop bit of received data is one. Always set this bit to zero when writing to UCSRA.

• Bit 3 – DOR: Data OverRun

This bit is set if a Data OverRun condition is detected. A Data OverRun occurs when the receive buffer is full (two characters), it is a new character waiting in the receive Shift Register, and a new start bit is detected. This bit is valid until the receive buffer (UDR) is read. Always set this bit to zero when writing to UCSRA.

• Bit 2 – PE: Parity Error

This bit is set if the next character in the receive buffer had a Parity Error when received and the parity checking was enabled at that point (UPM1 = 1). This bit is valid until the receive buffer (UDR) is read. Always set this bit to zero when writing to UCSRA.

• Bit 1 – U2X: Double the USART Transmission Speed

This bit only has effect for the asynchronous operation. Write this bit to zero when using synchronous operation.

Writing this bit to one will reduce the divisor of the baud rate divider from 16 to 8 effectively doubling the transfer rate for asynchronous communication.

USART Control and Status Register B – UCSRB

- **Bit 0 – MPCM: Multi-processor Communication Mode**

This bit enables the Multi-processor Communication mode. When the MPCM bit is written to one, all the incoming frames received by the USART receiver that do not contain address information will be ignored. The transmitter is unaffected by the MPCM setting. For more detailed information see [“Multi-processor Communication Mode” on page 161](#).

Bit	7	6	5	4	3	2	1	0	
	RXCIE	TXCIE	UDRIE	RXEN	TXEN	UCSZ2	RXB8	TXB8	UCSRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **Bit 7 – RXCIE: RX Complete Interrupt Enable**

Writing this bit to one enables interrupt on the RXC Flag. A USART Receive Complete Interrupt will be generated only if the RXCIE bit is written to one, the Global Interrupt Flag in SREG is written to one and the RXC bit in UCSRA is set.

- **Bit 6 – TXCIE: TX Complete Interrupt Enable**

Writing this bit to one enables interrupt on the TXC Flag. A USART Transmit Complete Interrupt will be generated only if the TXCIE bit is written to one, the Global Interrupt Flag in SREG is written to one and the TXC bit in UCSRA is set.

- **Bit 5 – UDRIE: USART Data Register Empty Interrupt Enable**

Writing this bit to one enables interrupt on the UDRE Flag. A Data Register Empty Interrupt will be generated only if the UDRIE bit is written to one, the Global Interrupt Flag in SREG is written to one and the UDRE bit in UCSRA is set.

- **Bit 4 – RXEN: Receiver Enable**

Writing this bit to one enables the USART Receiver. The Receiver will override normal port operation for the RxD pin when enabled. Disabling the Receiver will flush the receive buffer invalidating the FE, DOR, and PE Flags.

- **Bit 3 – TXEN: Transmitter Enable**

Writing this bit to one enables the USART Transmitter. The Transmitter will override normal port operation for the TxD pin when enabled. The disabling of the Transmitter (writing TXEN to zero) will not become effective until ongoing and pending transmissions are completed, that is, when the transmit Shift Register and transmit Buffer Register do not contain data to be transmitted. When disabled, the transmitter will no longer override the TxD port.

- **Bit 2 – UCSZ2: Character Size**

The UCSZ2 bits combined with the UCSZ1:0 bit in UCSRC sets the number of data bits (Character Size) in a frame the receiver and transmitter use.

- **Bit 1 – RXB8: Receive Data Bit 8**

RXB8 is the ninth data bit of the received character when operating with serial frames with nine data bits. Must be read before reading the low bits from UDR.

USART Control and Status Register C – UCSRC

• Bit 0 – TXB8: Transmit Data Bit 8

TXB8 is the ninth data bit in the character to be transmitted when operating with serial frames with nine data bits. Must be written before writing the low bits to UDR.

Bit	7	6	5	4	3	2	1	0	
	URSEL	UMSEL	UPM1	UPM0	USBS	UCSZ1	UCSZ0	UCPOL	UCSRC
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	1	0	0	0	0	1	1	0	

The UCSRC Register shares the same I/O location as the UBRRH Register. See the [“Accessing UBRRH/ UCSRC Registers” on page 162](#) section which describes how to access this register.

• Bit 7 – URSEL: Register Select

This bit selects between accessing the UCSRC or the UBRRH Register. It is read as one when reading UCSRC. The URSEL must be one when writing the UCSRC.

• Bit 6 – UMSEL: USART Mode Select

This bit selects between Asynchronous and Synchronous mode of operation.

Table 63. UMSEL Bit Settings

UMSEL	Mode
0	Asynchronous Operation
1	Synchronous Operation

• Bit 5:4 – UPM1:0: Parity Mode

These bits enable and set type of parity generation and check. If enabled, the transmitter will automatically generate and send the parity of the transmitted data bits within each frame. The Receiver will generate a parity value for the incoming data and compare it to the UPM0 setting. If a mismatch is detected, the PE Flag in UCSRA will be set.

Table 64. UPM Bits Settings

UPM1	UPM0	Parity Mode
0	0	Disabled
0	1	Reserved
1	0	Enabled, Even Parity
1	1	Enabled, Odd Parity

• Bit 3 – USBS: Stop Bit Select

This bit selects the number of Stop Bits to be inserted by the Transmitter. The Receiver ignores this setting.

Table 65. USBS Bit Settings

USBS	Stop Bit(s)
0	1-bit
1	2-bit

• Bit 2:1 – UCSZ1:0: Character Size

The UCSZ1:0 bits combined with the UCSZ2 bit in UCSRB sets the number of data bits (Character Size) in a frame the Receiver and Transmitter use.

Table 66. UCSZ Bits Settings

UCSZ2	UCSZ1	UCSZ0	Character Size
0	0	0	5-bit
0	0	1	6-bit
0	1	0	7-bit
0	1	1	8-bit
1	0	0	Reserved
1	0	1	Reserved
1	1	0	Reserved
1	1	1	9-bit

• Bit 0 – UCPOL: Clock Polarity

This bit is used for Synchronous mode only. Write this bit to zero when Asynchronous mode is used. The UCPOL bit sets the relationship between data output change and data input sample, and the synchronous clock (XCK).

Table 67. UCPOL Bit Settings

UCPOL	Transmitted Data Changed (Output of TxD Pin)	Received Data Sampled (Input on RxD Pin)
0	Rising XCK Edge	Falling XCK Edge
1	Falling XCK Edge	Rising XCK Edge

USART Baud Rate Registers – UBRRL and UBRRH

Bit	15	14	13	12	11	10	9	8	
	URSEL	–	–	–	UBRR[11:8]				UBRRH
	UBRR[7:0]								UBRRL
	7	6	5	4	3	2	1	0	
Read/Write	R/W	R	R	R	R/W	R/W	R/W	R/W	
	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	
	0	0	0	0	0	0	0	0	

The UBRRH Register shares the same I/O location as the UCSRC Register. See the [“Accessing UBRRH/ UCSRC Registers” on page 162](#) section which describes how to access this register.

• Bit 15 – URSEL: Register Select

This bit selects between accessing the UBRRH or the UCSRC Register. It is read as zero when reading UBRRH. The URSEL must be zero when writing the UBRRH.

• Bit 14:12 – Reserved Bits

These bits are reserved for future use. For compatibility with future devices, these bit must be written to zero when UBRRH is written.

• Bit 11:0 – UBRR11:0: USART Baud Rate Register

This is a 12-bit register which contains the USART baud rate. The UBRRH contains the four most significant bits, and the UBRL contains the 8 least significant bits of the USART baud rate. Ongoing transmissions by the transmitter and receiver will be corrupted if the baud rate is changed. Writing UBRL will trigger an immediate update of the baud rate prescaler.

Examples of Baud Rate Setting

For standard crystal and resonator frequencies, the most commonly used baud rates for asynchronous operation can be generated by using the UBRR settings in [Table 68](#). UBRR values which yield an actual baud rate differing less than 0.5% from the target baud rate, are bold in the table. Higher error ratings are acceptable, but the receiver will have less noise resistance when the error ratings are high, especially for large serial frames (see [“Asynchronous Operational Range” on page 159](#)). The error values are calculated using the following equation:

$$\text{Error}[\%] = \left(\frac{\text{BaudRate}_{\text{Closest Match}}}{\text{BaudRate}} - 1 \right) \cdot 100\%$$

Table 68. Examples of UBRR Settings for Commonly Used Oscillator Frequencies

Baud Rate (bps)	$f_{\text{osc}} = 1.0000 \text{ MHz}$				$f_{\text{osc}} = 1.8432 \text{ MHz}$				$f_{\text{osc}} = 2.0000 \text{ MHz}$			
	U2X = 0		U2X = 1		U2X = 0		U2X = 1		U2X = 0		U2X = 1	
	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error
2400	25	0.2%	51	0.2%	47	0.0%	95	0.0%	51	0.2%	103	0.2%
4800	12	0.2%	25	0.2%	23	0.0%	47	0.0%	25	0.2%	51	0.2%
9600	6	-7.0%	12	0.2%	11	0.0%	23	0.0%	12	0.2%	25	0.2%
14.4k	3	8.5%	8	-3.5%	7	0.0%	15	0.0%	8	-3.5%	16	2.1%
19.2k	2	8.5%	6	-7.0%	5	0.0%	11	0.0%	6	-7.0%	12	0.2%
28.8k	1	8.5%	3	8.5%	3	0.0%	7	0.0%	3	8.5%	8	-3.5%
38.4k	1	-18.6%	2	8.5%	2	0.0%	5	0.0%	2	8.5%	6	-7.0%
57.6k	0	8.5%	1	8.5%	1	0.0%	3	0.0%	1	8.5%	3	8.5%
76.8k	–	–	1	-18.6%	1	-25.0%	2	0.0%	1	-18.6%	2	8.5%
115.2k	–	–	0	8.5%	0	0.0%	1	0.0%	0	8.5%	1	8.5%
230.4k	–	–	–	–	–	–	0	0.0%	–	–	–	–
250k	–	–	–	–	–	–	–	–	–	–	0	0.0%
Max ⁽¹⁾	62.5 Kbps		125 Kbps		115.2 Kbps		230.4 Kbps		125 Kbps		250 Kbps	

1. UBRR = 0, Error = 0.0%

Table 69. Examples of UBRR Settings for Commonly Used Oscillator Frequencies (Continued)

Baud Rate (bps)	$f_{osc} = 3.6864 \text{ MHz}$				$f_{osc} = 4.0000 \text{ MHz}$				$f_{osc} = 7.3728 \text{ MHz}$			
	U2X = 0		U2X = 1		U2X = 0		U2X = 1		U2X = 0		U2X = 1	
	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error
2400	95	0.0%	191	0.0%	103	0.2%	207	0.2%	191	0.0%	383	0.0%
4800	47	0.0%	95	0.0%	51	0.2%	103	0.2%	95	0.0%	191	0.0%
9600	23	0.0%	47	0.0%	25	0.2%	51	0.2%	47	0.0%	95	0.0%
14.4k	15	0.0%	31	0.0%	16	2.1%	34	-0.8%	31	0.0%	63	0.0%
19.2k	11	0.0%	23	0.0%	12	0.2%	25	0.2%	23	0.0%	47	0.0%
28.8k	7	0.0%	15	0.0%	8	-3.5%	16	2.1%	15	0.0%	31	0.0%
38.4k	5	0.0%	11	0.0%	6	-7.0%	12	0.2%	11	0.0%	23	0.0%
57.6k	3	0.0%	7	0.0%	3	8.5%	8	-3.5%	7	0.0%	15	0.0%
76.8k	2	0.0%	5	0.0%	2	8.5%	6	-7.0%	5	0.0%	11	0.0%
115.2k	1	0.0%	3	0.0%	1	8.5%	3	8.5%	3	0.0%	7	0.0%
230.4k	0	0.0%	1	0.0%	0	8.5%	1	8.5%	1	0.0%	3	0.0%
250k	0	-7.8%	1	-7.8%	0	0.0%	1	0.0%	1	-7.8%	3	-7.8%
0.5M	—	—	0	-7.8%	—	—	0	0.0%	0	-7.8%	1	-7.8%
1M	—	—	—	—	—	—	—	—	—	—	0	-7.8%
Max ⁽¹⁾	230.4 Kbps		460.8 Kbps		250 Kbps		0.5 Mbps		460.8 Kbps		921.6 Kbps	

1. UBRR = 0, Error = 0.0%

Table 70. Examples of UBRR Settings for Commonly Used Oscillator Frequencies (Continued)

Baud Rate (bps)	$f_{osc} = 8.0000 \text{ MHz}$				$f_{osc} = 11.0592 \text{ MHz}$				$f_{osc} = 14.7456 \text{ MHz}$			
	U2X = 0		U2X = 1		U2X = 0		U2X = 1		U2X = 0		U2X = 1	
	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error
2400	207	0.2%	416	-0.1%	287	0.0%	575	0.0%	383	0.0%	767	0.0%
4800	103	0.2%	207	0.2%	143	0.0%	287	0.0%	191	0.0%	383	0.0%
9600	51	0.2%	103	0.2%	71	0.0%	143	0.0%	95	0.0%	191	0.0%
14.4k	34	-0.8%	68	0.6%	47	0.0%	95	0.0%	63	0.0%	127	0.0%
19.2k	25	0.2%	51	0.2%	35	0.0%	71	0.0%	47	0.0%	95	0.0%
28.8k	16	2.1%	34	-0.8%	23	0.0%	47	0.0%	31	0.0%	63	0.0%
38.4k	12	0.2%	25	0.2%	17	0.0%	35	0.0%	23	0.0%	47	0.0%
57.6k	8	-3.5%	16	2.1%	11	0.0%	23	0.0%	15	0.0%	31	0.0%
76.8k	6	-7.0%	12	0.2%	8	0.0%	17	0.0%	11	0.0%	23	0.0%
115.2k	3	8.5%	8	-3.5%	5	0.0%	11	0.0%	7	0.0%	15	0.0%
230.4k	1	8.5%	3	8.5%	2	0.0%	5	0.0%	3	0.0%	7	0.0%
250k	1	0.0%	3	0.0%	2	-7.8%	5	-7.8%	3	-7.8%	6	5.3%
0.5M	0	0.0%	1	0.0%	—	—	2	-7.8%	1	-7.8%	3	-7.8%
1M	—	—	0	0.0%	—	—	—	—	0	-7.8%	1	-7.8%
Max ⁽¹⁾	0.5 Mbps		1 Mbps		691.2 Kbps		1.3824 Mbps		921.6 Kbps		1.8432 Mbps	

1. UBRR = 0, Error = 0.0%

Table 71. Examples of UBRR Settings for Commonly Used Oscillator Frequencies (Continued)

Baud Rate (bps)	$f_{osc} = 16.0000 \text{ MHz}$				$f_{osc} = 18.4320 \text{ MHz}$				$f_{osc} = 20.0000 \text{ MHz}$			
	U2X = 0		U2X = 1		U2X = 0		U2X = 1		U2X = 0		U2X = 1	
	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error
2400	416	-0.1%	832	0.0%	479	0.0%	959	0.0%	520	0.0%	1041	0.0%
4800	207	0.2%	416	-0.1%	239	0.0%	479	0.0%	259	0.2%	520	0.0%
9600	103	0.2%	207	0.2%	119	0.0%	239	0.0%	129	0.2%	259	0.2%
14.4k	68	0.6%	138	-0.1%	79	0.0%	159	0.0%	86	-0.2%	173	-0.2%
19.2k	51	0.2%	103	0.2%	59	0.0%	119	0.0%	64	0.2%	129	0.2%
28.8k	34	-0.8%	68	0.6%	39	0.0%	79	0.0%	42	0.9%	86	-0.2%
38.4k	25	0.2%	51	0.2%	29	0.0%	59	0.0%	32	-1.4%	64	0.2%
57.6k	16	2.1%	34	-0.8%	19	0.0%	39	0.0%	21	-1.4%	42	0.9%
76.8k	12	0.2%	25	0.2%	14	0.0%	29	0.0%	15	1.7%	32	-1.4%
115.2k	8	-3.5%	16	2.1%	9	0.0%	19	0.0%	10	-1.4%	21	-1.4%
230.4k	3	8.5%	8	-3.5%	4	0.0%	9	0.0%	4	8.5%	10	-1.4%
250k	3	0.0%	7	0.0%	4	-7.8%	8	2.4%	4	0.0%	9	0.0%
0.5M	1	0.0%	3	0.0%	—	—	4	-7.8%	—	—	4	0.0%
1M	0	0.0%	1	0.0%	—	—	—	—	—	—	—	—
Max ⁽¹⁾	1 Mbps		2 Mbps		1.152 Mbps		2.304 Mbps		1.25 Mbps		2.5 Mbps	

1. UBRR = 0, Error = 0.0%